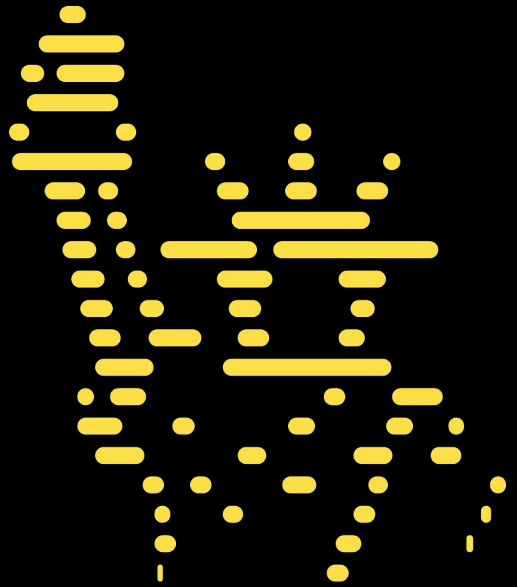
aiDevCon
by Tessl

NYC

Scaling Agentic Engineering at Zalando

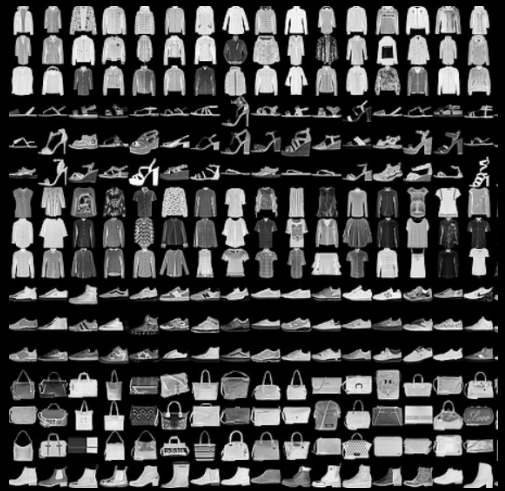


Bartosz Ocytko
Executive Principal Engineer · Zalando

Zalando - Open Source

Creators of **patroni**
PostgreSQL HA

postgres-operator



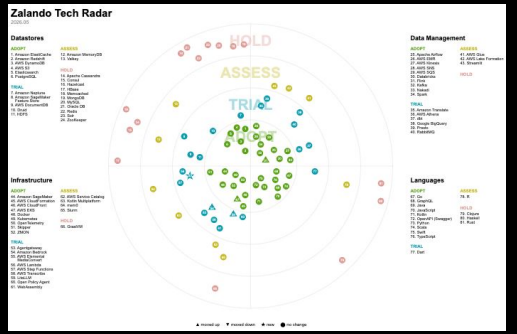
Fashion-MNIST



skipper
HTTP router,
Kubernetes Ingress



restful-api-guidelines



Zalando tech-radar

agenticidentitybroker.dev
Governed delegated access for AI agents,
RFC 8693

Zalando - The leading European technology platform for fashion and lifestyle.



>60m
active customers
(Zalando and About You)

€17.6bn
GMV in 2025

27
markets in Europe

250+
teams

2000
developers

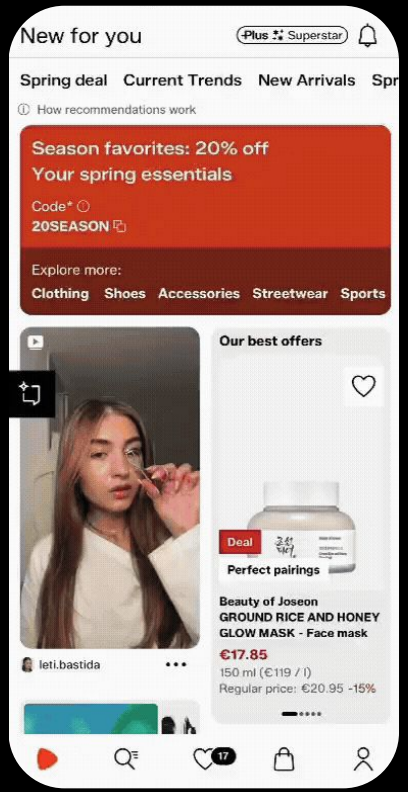
300+
AWS accounts

250+
Kubernetes clusters

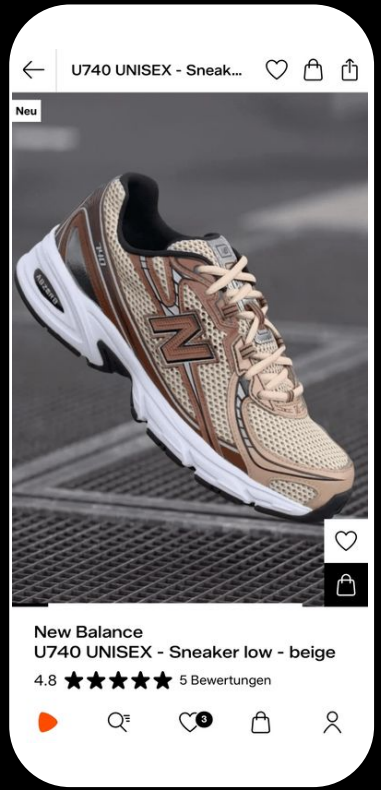
2700+ applications
(Java, Kotlin, Scala, Go, Python)

Zalando - The leading European technology platform for fashion and lifestyle.

Home Feed



Elevated Content



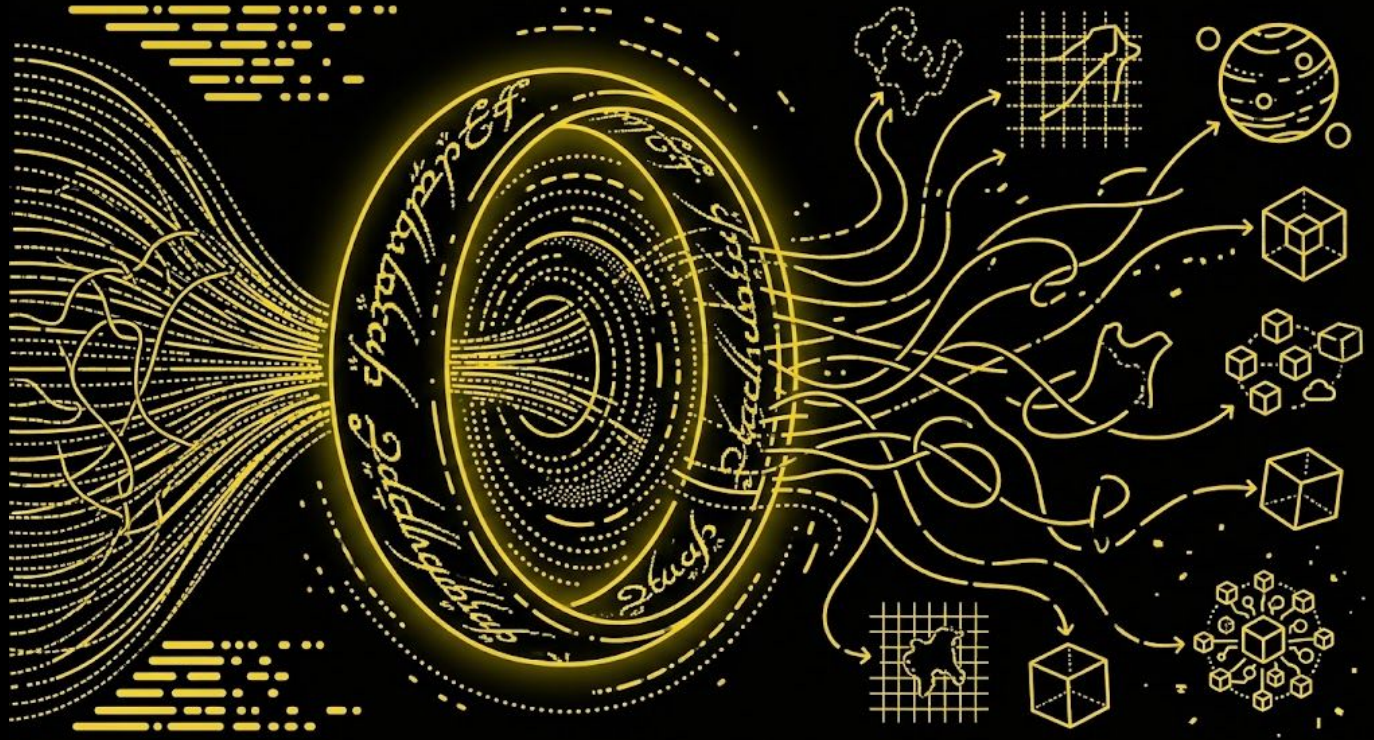
Personalised inspiration: Zalando Assistant

22m

unique users this year



One proxy to rule ‘em all



One proxy to rule ‘em all

- central entry point for all Builders
- vendor & model independence
- tool flexibility
- control the tool sprawl

LiteLLM: to love it or to hate it?!



Extensibility

- anonymized usage stats
- forced prompt caching
- forced client upgrades
- backporting fixes

Stability

--max_requests_before_restart = 20000

- read blog & release notes
- avoid DB if you can
- upgrade often enough

bedrock/anthropic.claude-opus-5-5	in: 0.002M	cache↓: 1.542M	cache↑: 1.421M	out: 0.025M	total: 2.989M	out: 65.2 tok/s	cost: $7.9160
gpt-6.1-sol	in: 0.000M	cache↓: 1.353M	cache↑: 0.140M	out: 0.032M	total: 1.524M	out: 60.3 tok/s	cost: $0.8053
gpt-6.1-terra	in: 0.002M	cache↓: 0.001M	cache↑: 0.000M	out: 0.002M	total: 0.006M	out: 79.4 tok/s	cost: $0.0143
gpt-6-luna	in: 0.000M	cache↓: 0.000M	cache↑: 0.000M	out: 0.000M	total: 0.000M	out: 8.7 tok/s	cost: $0.0000
zllm total	in: 0.005M	cache↓: 2.896M	cache↑: 1.560M	out: 0.059M	total: 4.520M	out: 62.9 tok/s	cost: $8.7356

zllm-cli

- pre claude code
- used in trainings
- auth injection to avoid hardcoding secrets for MCP
- configure coding agents

plugins

- opencode, pi model providers
- reasoning effort param mapping fixes
- file-based fallback cache

zllm-proxy

- Authorization header injection for stubborn tools (no apiKeyHelper)
- request/response logging
- prompt cache warnings

Impact on code



Leverage in platform teams

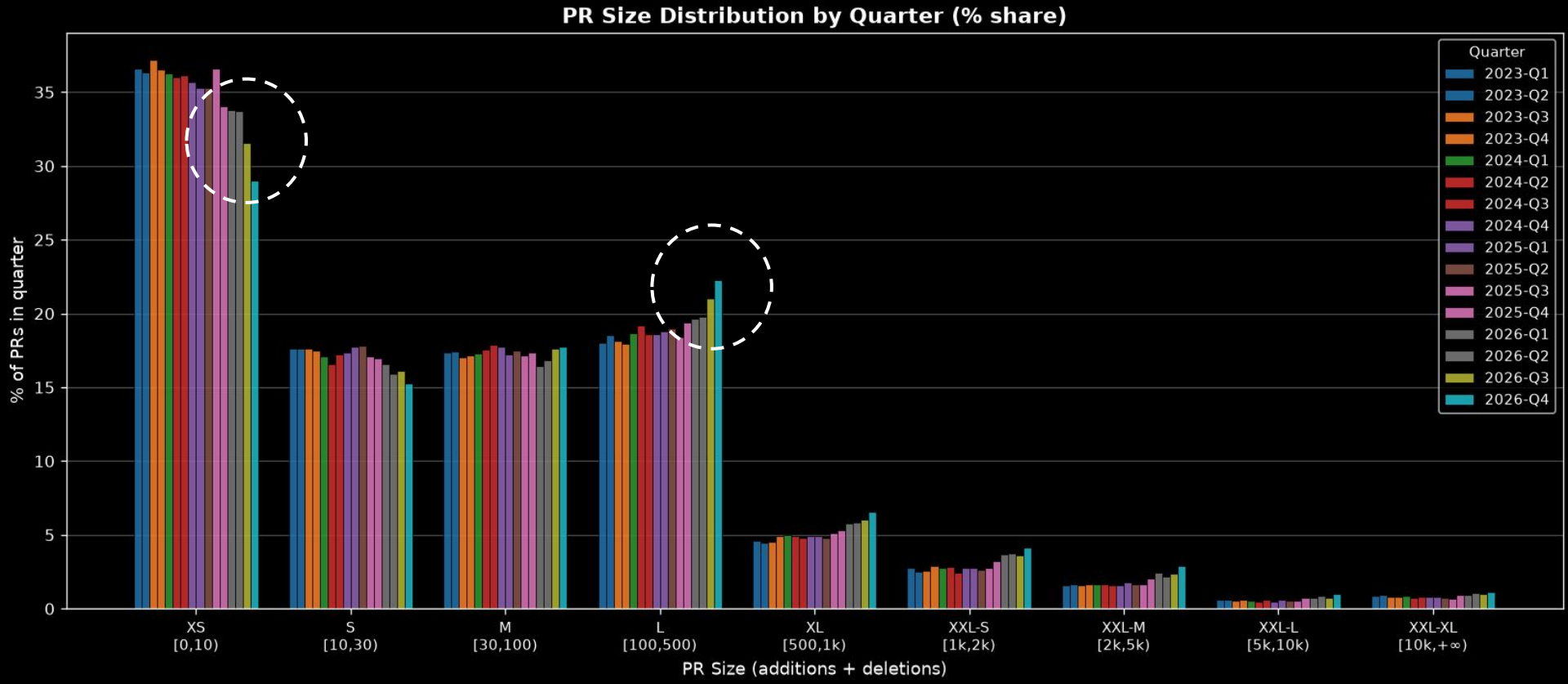
90%
acceptance rate

for automated PRs across 1000s+ repositories

LLM-based transformations help platform teams cover more ground

44% lower effort on migrations

migration owner takes full ownership
for >90% migration scope


PR Size Distribution by Quarter (% share)
% of PRs in quarter
0
5
10
15
20
25
30
35
XS
[0,10)
S
[10,30)
M
[30,100)
L
[100,500)
XL
[500,1k)
XXL-S
[1k,2k)
XXL-M
[2k,5k)
XXL-L
[5k,10k)
XXL-XL
[10k,+∞)
PR Size (additions + deletions)
Quarter
2023-Q1
2023-Q2
2023-Q3
2023-Q4
2024-Q1
2024-Q2
2024-Q3
2024-Q4
2025-Q1
2025-Q2
2025-Q3
2025-Q4
2026-Q1
2026-Q2
2026-Q3
2026-Q4

Risk-based PR scoring

>33% PRs are low risk

20-40% lower lead time
if merged by author

Risk-based PR scoring



4. Index creation without CONCURRENTLY in V1_23

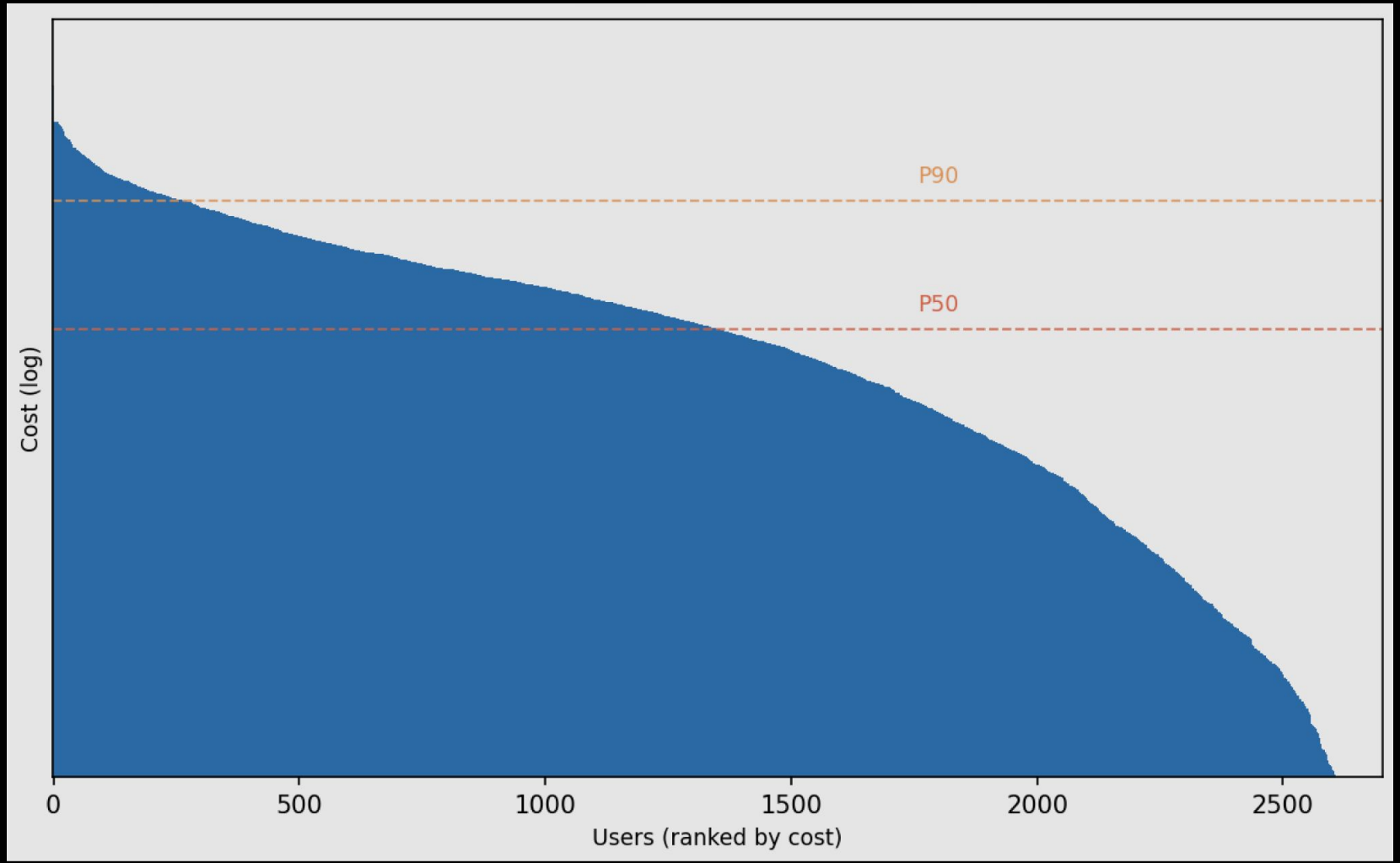
The `CREATE INDEX` in V1_23 runs in a normal transaction (takes `ShareLock` on the table). The README documents a registration pause to mitigate this, but any concurrent reads during migration will be blocked. This is an operational risk if the pause is not perfectly coordinated.

5. Complex, multi-stage operational upgrade

The upgrade requires pausing registration, staged Flyway migration, manual artifact verification queries, and coordinated rollout of all readers/writers. Any deviation from the documented sequence risks data inconsistency or service unavailability.

Guardrails & Governance

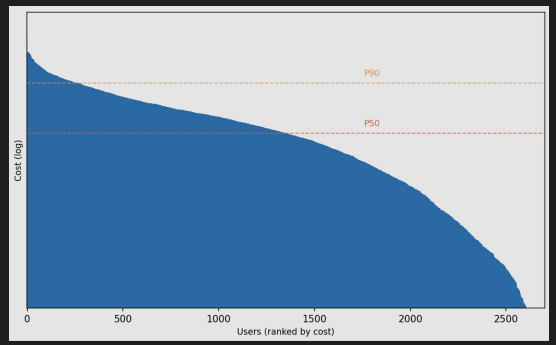



P90
P50
Cost (log)
0
500
1000
1500
2000
2500
Users (ranked by cost)

Guardrails

Observe

p90 cost/user
is >10x the p50



Educate

top spenders
learn from power users

local cost view
agentsview & codeburn

trainings

Protect

cost/user
budgeting

lead time & CFR
as guardrail DORA metrics

notifications & spend cap
stop runaway cost growth

Governance

Govern

Tech Radar

OSS Licensing Guide

AI systems & models
in developer portal

Support

self-assessment
legal scorecards

legal review for high-risk use cases

production readiness reviews

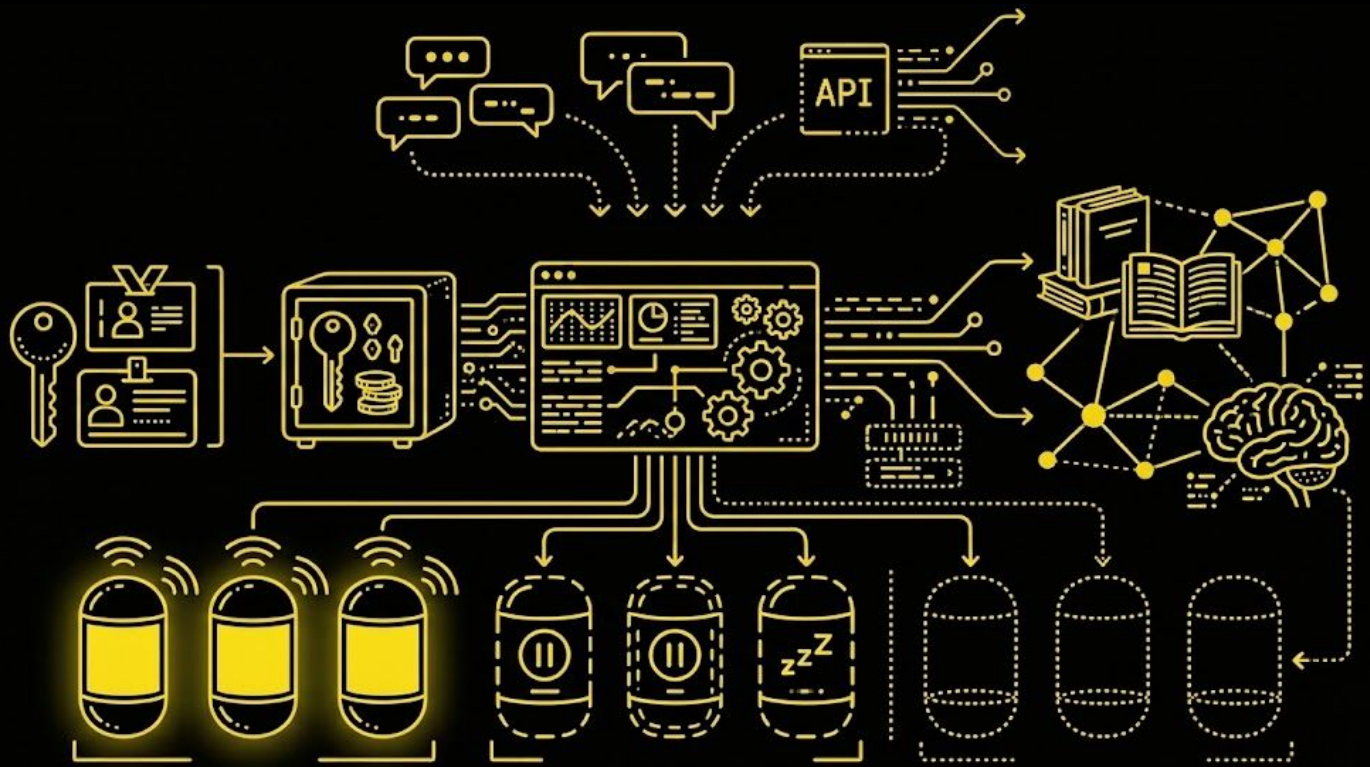
Verify

AI model usage detection
docker image scan

offline & online evals

E2E probes in prod

Zalando Agent Platform



Zalando Agent Platform

Goals

hosted agents
use Kubernetes resource slack

external MCP servers

governance
MCP servers, skill marketplaces

Foundations

kagent
runtime

agentgateway
model access, routing

identity broker
secure on-behalf-of flows

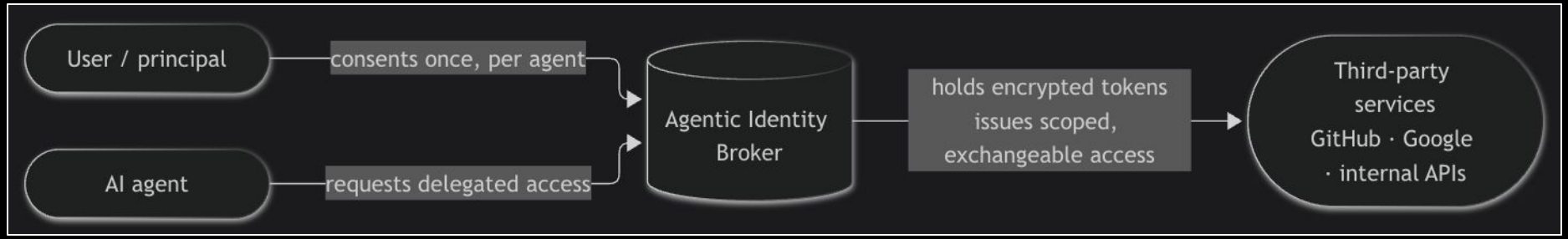
Add-ons

caura
memory

sharing agents
promote re-use

agent templates
quick startup

agenticidentitybroker



- Identity infrastructure for agents that act on a user's behalf
- Token vault + Token exchange at the edge, RFC8693
- OSS by Zalando (agenticidentitybroker.dev), Golang, spec-driven

Creating an agent

Guidance updates as you move through the setup.

PROMPT

Define its behavior

The prompt is the agent's standing instruction for every conversation.

Prompt

State the agent's purpose, expected tone, boundaries, and how it should use available tools. Include useful defaults and escalation guidance.

Memory

Choose whether the agent and its subagents remember things between conversations: not at all, privately per user, or shared across all users of the agent.

New Agent

Prompt

Basic info › Icon › Prompt › MCPs › Review

Prompt

This prompt defines the agent's behavior for every conversation.

Prompt *

```
You are a meeting scheduling assistant. You have
three capabilities:

1. Given a list of names, you find when they have
free time to meet for a meeting. Max. lookahead is
1 month in advance (unless specified otherwise).
Default meeting time is 30 minutes. You use the
Sunrise MCP to look up team names, resolve email
addresses from names, etc.
2. You offer scheduling the meeting and ask user to
confirm the time AND set the meeting title and
agenda.
3. Set preferences on preferred meeting times and
focus times for the current user and store them in
the agent memory.

ALWAYS ASK FOR CONFIRMATION BEFORE SENDING MESSAGES
OR INVITES.
```

Memory

Choose whether this agent and its subagents remember things between conversations.

None | Private | Shared

Memory is kept separately for each user and is never shared with other users of the agent.

New Agent

MCPs

Basic info › Icon › Prompt › MCPs › Review

MCPs

Select the MCPProxy resources this agent may use. References are stored by Kubernetes resource name and namespace.

Sunrise MCP × Google Gmail & Calendar ×

Search MCPs...

STARRED

- [ ] **Linear MCP**
  Proxy fronting the Linear MCP backend.
- [x] **Sunrise MCP**
  Proxy fronting the Sunrise MCP.
- [ ] **Ziri MCP**
  Ziri provides context about internal knowledge of various sources like Google Docs, Google Drive, Mkdocs, Confluence, Github Readmes etc.

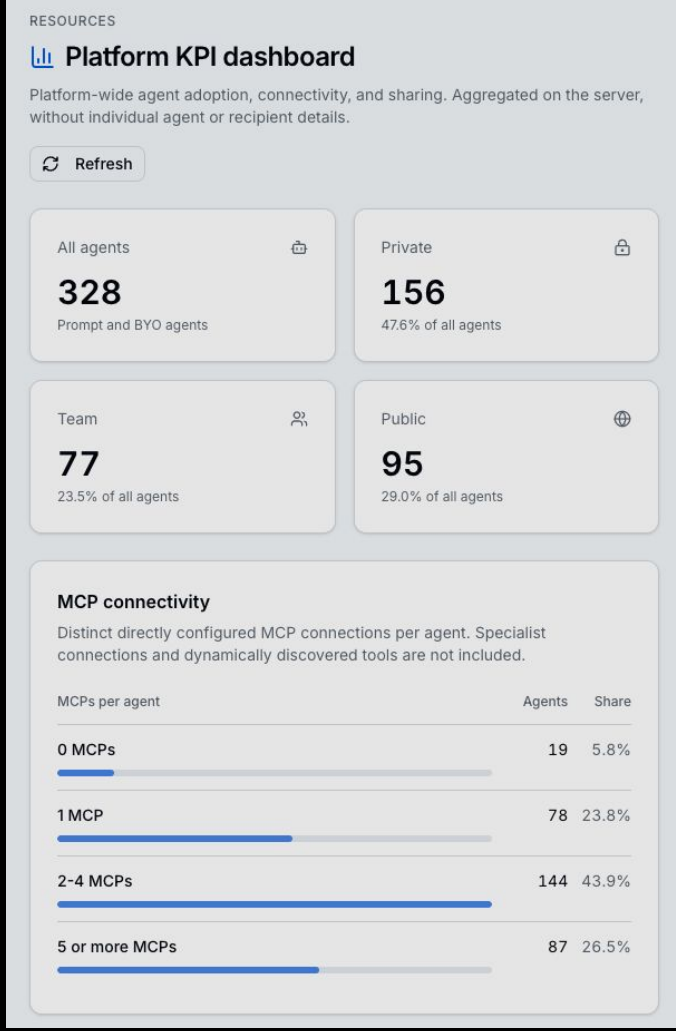
Agent resources





Zalando Agent Platform

- Agent access: API, Chat UI, Google Chat
- >300 agents (in 2 weeks)
- Agent deduplication?
- Review for Agents?



Citizen developers



Citizen developers

- Gated by tooling access
- Never onboarded as developers
- Make AI-assisted PRs
- Let the agent solve “everything”
- Need help in productizing tools

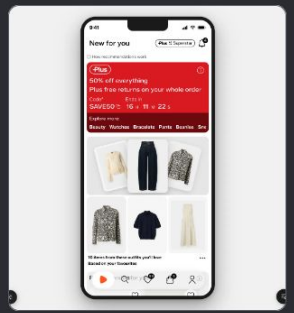
Templates

A complete, reusable page layout intended to be duplicated and adapted.

By premise

- Home
- Catalog
- Wishlist
- PDP
- Cart
- Checkout
- User Account
- Boards
- Outfits
- ZA
- Video & Livestream

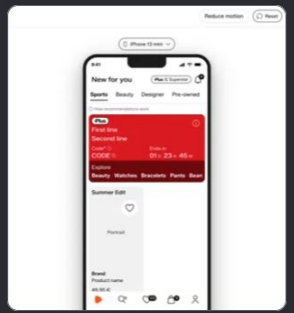
Recently added



Feed Large Card - App

Single-column home feed using the Large Base Card — same content as the small-card feed but full-width cards with inverted anatomy (profile above asset, H5 titles below).

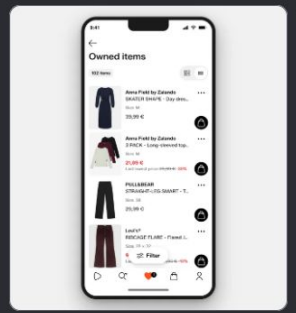
Oct 6 • Thiago Tomasi



Feed - App

Home feed with Salutation, category navigation, Sales Banner, and bottom navigation.

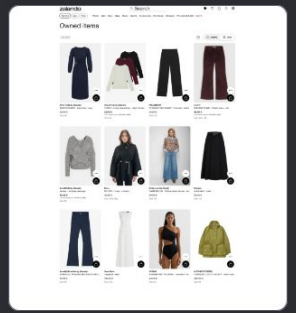
Sep 30 • Thiago Tomasi



Owned Items - App

Mobile owned-items list with item status, prices, sizes, and quick product actions.

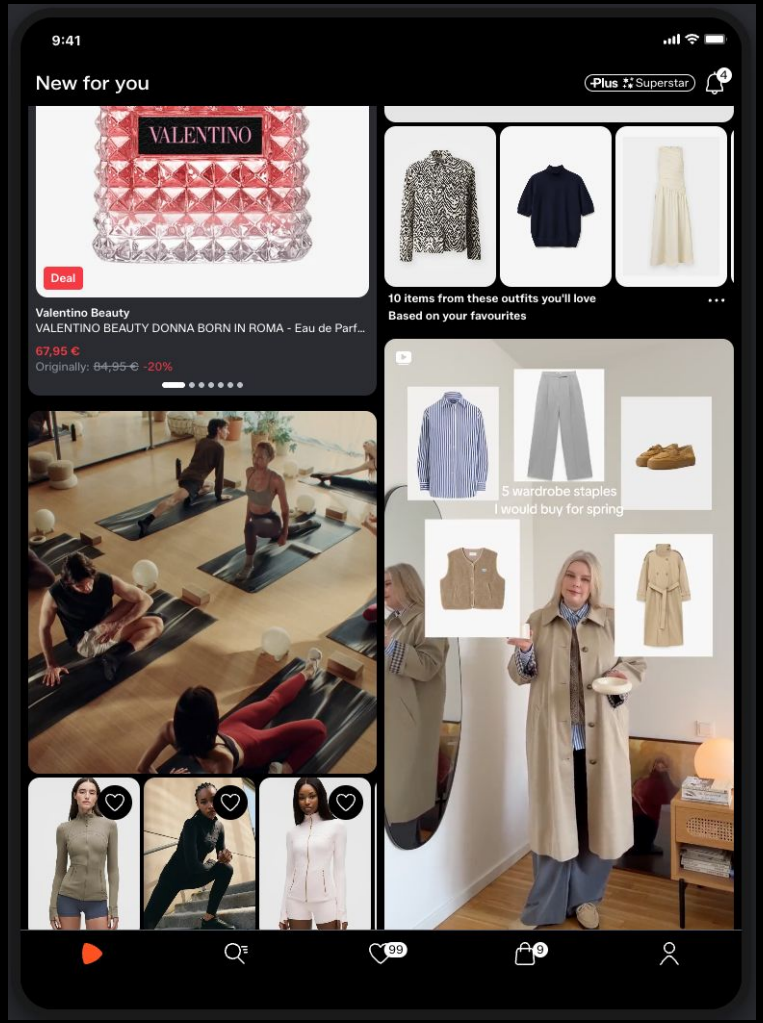
Sep 29 • Tiago Ramos



Owned Items - Desktop

Desktop owned-items board with responsive product views, item states, prices, sizes, and wardrobe actions.

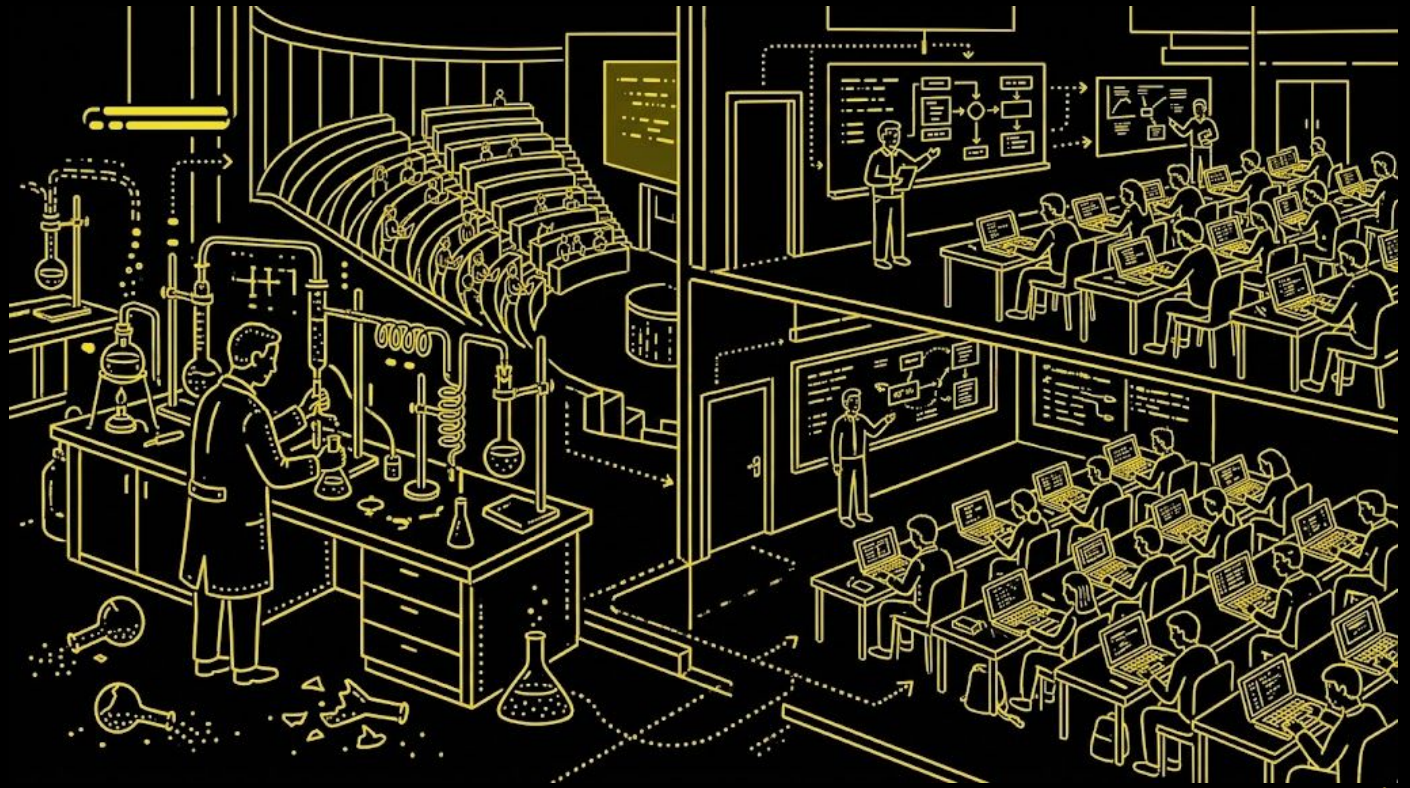
Sep 29 • Tiago Ramos



Productizing needs

- **Repo + skills**
  => App with Chat UI
- **Tool + prototype configs in one repo**
  => separate publishing flow
- **Internal only**
  => external for user-research

Enablement



Developers' challenges

- Pace is too high to keep up
- Insufficient time to learn
- Feel relief after peer exchange
- Unclear outlook for 2-3 years

Our enablement

- #guild-LLMs
- 4-hour labs => trainings
- 4-week training
- AI readiness scan for repos

“Not everything that *can* be built,
should be built”

aiDevCon
by Tessl

NYC

Bartosz Ocytko

🐘 @bocytko (X, hachyderm.io)
📝 ocytko.net



📝 engineering.zalando.com
🌍 agenticidentitybroker.dev

aiDevCon

by Tessl

NYC



November 2-4

New York and virtual